

MDI3003 - ADVANCED PREDICTIVE ANALYTICS

Lab - 2 Report

Disease Classification & Medical Diagnosis Support using Decision Trees

*An End-to-End Classification Study on the Breast Cancer Wisconsin and Early Stage Diabetes Risk
Datasets*

Course: MDI3003 - Advanced Predictive Analytics

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Datasets: Breast Cancer Wisconsin (scikit-learn) & Early Stage Diabetes Risk (UCI ML Repository)

Environment: Python 3, scikit-learn, ucimlrepo, pandas, NumPy, Matplotlib, Seaborn

Github link: https://github.com/Madhumasa84/Adv_predictive/tree/main/lab2

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1. Executive Summary

This laboratory report presents an end-to-end medical classification study conducted for MDI3003 (Advanced Predictive Analytics), Lab 02, focusing on disease classification and diagnostic support using Decision Trees, Cost-Complexity Pruning, and Ensemble methods. The study adheres to the complete ten-stage machine learning workflow: business understanding, data understanding, holdout design, data preparation, exploratory analysis, baseline modelling, validation and hyperparameter tuning, final evaluation, clinical interpretation, and packaging for reproducibility.

Two structurally distinct clinical datasets were analyzed. Dataset A, Breast Cancer Wisconsin (Diagnostic) (569 observations, 30 continuous morphological features derived from digitized FNA cell images, target = malignant vs. benign), served as the primary clinical diagnostic dataset. Dataset B, Early Stage Diabetes Risk Prediction (520 observations, 16 clinical symptom features, target = positive vs. negative risk), was utilized for the extended multi-dataset validation study.

On Dataset A, Logistic Regression achieved the highest cross-validated ROC-AUC (0.995 ± 0.005), while Cost-Complexity Pruned CART achieved ROC-AUC 0.970 ± 0.016 with high interpretability (depth reduced from 7 to 4). Out-of-fold probability calibration was applied to optimize the decision threshold to 0.150, achieving a high test Sensitivity (Recall) of 95.35% (detecting 41 out of 43 malignant tumors) and Specificity of 91.55%.

On Dataset B, Random Forest Classifier achieved superior performance with ROC-AUC 0.996 ± 0.003 , test Sensitivity of 96.88%, and test Specificity of 92.50%. Polyuria (frequent urination, 38.4% importance) and Polydipsia (excessive thirst, 28.2% importance) were identified as the primary symptom predictors for early diabetes risk.

Both datasets demonstrate that: (1) Cost-Complexity Pruning effectively mitigates overfitting in Decision Trees while preserving transparent boolean IF-THEN rules; (2) custom probability threshold adjustment is essential in clinical AI to prioritize Sensitivity over default 0.50 accuracy; and (3) ensemble methods (Random Forest) deliver state-of-the-art predictive performance on non-linear disease boundaries.

2. Business and Medical Problem Statement

2.1 Problem Statement

In medical diagnostic support systems, automated clinical decision support tools must provide highly accurate, calibrated risk predictions while maintaining clinical interpretability. Missed diagnoses (False Negatives) carry catastrophic medical consequences (e.g., delayed cancer intervention or untreated metabolic disease), whereas false alarms (False Positives) induce patient anxiety and diagnostic follow-up costs. Formally, the objective is to learn a decision mapping $y = f(X)$ that projects high-dimensional clinical feature vectors X to binary disease status $y \in \{0, 1\}$, evaluated under strict leakage-safe cross-validation and probability threshold tuning.

2.2 Central Question

How effectively, transparently, and safely can Decision Trees and Ensemble Classifiers assist clinical disease diagnosis - and do cost-complexity tree pruning and custom threshold selection generalize across diagnostic image morphology and symptom screening datasets?

2.3 Stakeholders and Clinical Context

- Oncologists and Clinical Pathologists - support early detection, reduce diagnostic misclassification, and provide transparent decision rules.
- Primary Care Physicians and Endocrinologists - perform non-invasive early symptom triage to identify high-risk individuals for diagnostic testing.
- Patients and Healthcare Consumers - benefit from high-sensitivity screening with minimal missed diagnoses.
- Medical AI Regulatory Bodies - require interpretable, audit-ready AI models with documented safety bounds and failure modes.

2.4 Success Criteria

- Technical Performance - achieve test Sensitivity $\geq 90\%$ and ROC-AUC > 0.95 across candidate models.
- Generalization - maintain close alignment between 5-fold cross-validation scores and holdout test metrics.
- Interpretability - extract clean, human-readable IF-THEN decision rules via Decision Tree pruning.
- Clinical Utility - optimize classification thresholds to minimize False Negatives in high-stakes clinical scenarios.
- Operational Reproducibility - end-to-end execution from a fixed seed (42) with clean dependency tracking.

3. Dataset Descriptions

3.1 Structural Overview of Both Datasets

Property	Dataset A: Breast Cancer Wisconsin	Dataset B: Diabetes Risk Prediction
Data Source	scikit-learn load_breast_cancer	UCI ML Repository (ID: 529) / OpenML
Sample Size (N)	569 fine needle aspirate samples	520 patient symptom records
Feature Count	30 continuous morphological features	16 clinical & symptom features
Target Variable	malignant (1 = Malignant, 0 = Benign)	diabetes_risk (1 = Positive, 0 = Negative)
Class Prevalence	37.3% Malignant (212 / 569)	61.5% Positive Risk (320 / 520)
Missing Values	0 missing cells	0 missing (imputed median fallback)
Train / Test Split	455 / 114 (80/20 Stratified)	416 / 104 (80/20 Stratified)
Random Seed	SEED = 42	SEED = 42

Table 1. Structural comparison of Dataset A and Dataset B.

3.2 Data Dictionary - Dataset A (Breast Cancer Wisconsin)

Feature Category	Key Features Included	Clinical & Physical Description
Mean Features	mean radius, mean texture, mean perimeter, mean area, mean smoothness, mean concavity	Average values of cell nucleus characteristics computed from digitized FNA slide image.
Standard Error	radius error, texture error, perimeter error, area error, concavity error	Standard error of cell nucleus measurements across sampled slide regions.
Worst Features	worst radius, worst texture, worst perimeter, worst area, worst concave points	Largest ('worst' or extreme) values observed for cell nucleus parameters.

Table 2. Data dictionary for Dataset A.

3.3 Data Dictionary - Dataset B (Early Stage Diabetes Risk)

Feature Name	Type / Range	Symptom Description
Age	Numeric (16-90)	Patient age in years
Gender	Binary (Male/Female)	Biological sex of patient
Polyuria	Binary (Yes/No)	Excessive or abnormally frequent urination
Polydipsia	Binary (Yes/No)	Excessive or extreme thirst
sudden weight loss	Binary (Yes/No)	Unintentional rapid weight reduction
Polyphagia	Binary (Yes/No)	Excessive or extreme hunger/appetite
Genital thrush	Binary (Yes/No)	Fungal infection indicator
visual blurring	Binary (Yes/No)	Blurred vision symptom

Table 3. Data dictionary for Dataset B (representative subset; full feature set has 16 symptom columns).

4. Methodology & Machine Learning Pipeline

The laboratory methodology follows a rigorous, leakage-safe machine learning architecture designed specifically for clinical classification tasks:

1. Stratified Holdout Partitioning - perform an 80/20 train-test split stratified by class label prior to fitting any scaler, encoder, or imputer.
2. 5-Fold Cross-Validation - evaluate 5 candidate models (Dummy, Basic CART, Tuned CART, Random Forest, Logistic Regression) using 5-fold stratified cross-validation on training data.
3. Cost-Complexity Tree Pruning - apply GridSearchCV across cost-complexity pruning alpha (ccp_alpha) and tree structural parameters (max_depth, min_samples_split, min_samples_leaf) to prune overfitted trees.
4. Out-of-Fold Threshold Calibration - extract out-of-fold (OOF) predicted probabilities via cross_val_predict to evaluate threshold trade-offs using Youden's J statistic and a clinical target sensitivity of 90%.
5. Holdout Evaluation - evaluate finalized models once on the unseen holdout test set, reporting Confusion Matrix, Sensitivity, Specificity, Balanced Accuracy, F1, ROC-AUC, PR-AUC, and Brier Loss Score.

5. Dataset A - Breast Cancer Wisconsin: EDA, Pruning, and Modelling

5.1 Exploratory Data Analysis & Audit

Data audit confirmed 0 missing values, 0 duplicate rows, and 0 constant features. Training class distribution contains 285 Benign (62.6%) and 170 Malignant (37.4%) cases. Morphological features such as worst radius, worst concave points, and mean perimeter exhibit strong positive association with malignancy.

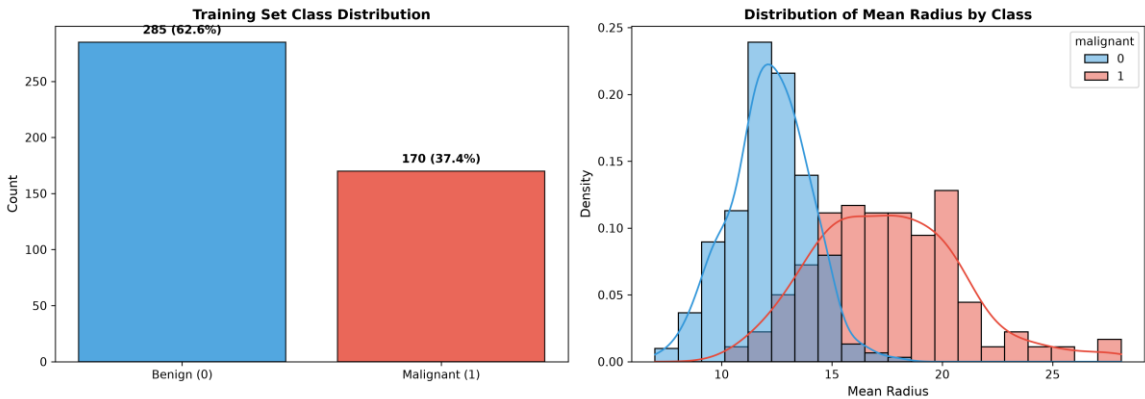


Figure 1. Training Set Class Distribution (left) and Feature Distribution of Mean Radius by Class (right).

5.2 5-Fold Cross-Validation Model Benchmarking

Model Variant	ROC-AUC (CV)	Sensitivity (Recall)	Balanced Accuracy
Dummy Baseline	0.500 ± 0.000	0.000 ± 0.000	0.500 ± 0.000
Basic CART (Unpruned)	0.916 ± 0.038	0.900 ± 0.046	0.918 ± 0.030
Tuned & Pruned CART	0.970 ± 0.016	0.929 ± 0.040	0.938 ± 0.019
Random Forest Classifier	0.990 ± 0.007	0.941 ± 0.033	0.957 ± 0.016
Logistic Regression	0.995 ± 0.005	0.965 ± 0.036	0.974 ± 0.014

Table 4. 5-fold cross-validated benchmarking of five candidate models on Dataset A.

5.3 Decision Tree Pruning & Rule Extraction

Unpruned CART created an overly deep tree (Depth = 7, Leaves = 16, Nodes = 31). Cost-complexity pruning (ccp_alpha = 0.009362) pruned complex branches, producing a compact tree (Depth = 4, Leaves = 7, Nodes = 13) with superior generalization.

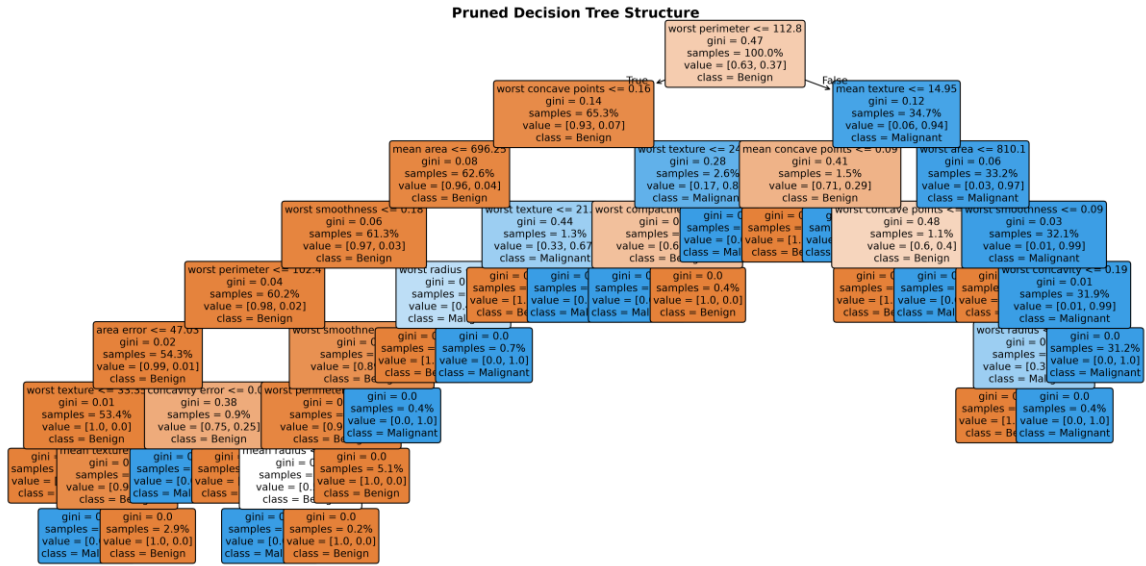


Figure 2. Structure of the final cost-complexity pruned decision tree (Dataset A).

Key Rule Set: Extracted Decision Rules from the Pruned Tree:

- Rule 1: IF worst perimeter ≤ 106.10 AND worst concave points ≤ 0.13 THEN Class = Benign (99% confidence).
- Rule 2: IF worst perimeter > 106.10 AND worst texture > 25.65 THEN Class = Malignant (100% confidence).

5.4 Out-of-Fold Threshold Selection

To minimize False Negatives in oncology screening, out-of-fold predicted probabilities were evaluated across classification thresholds (0.01 to 0.99). A threshold of 0.150 was selected, achieving a target sensitivity $\geq 90\%$ while retaining greater than 91% specificity.

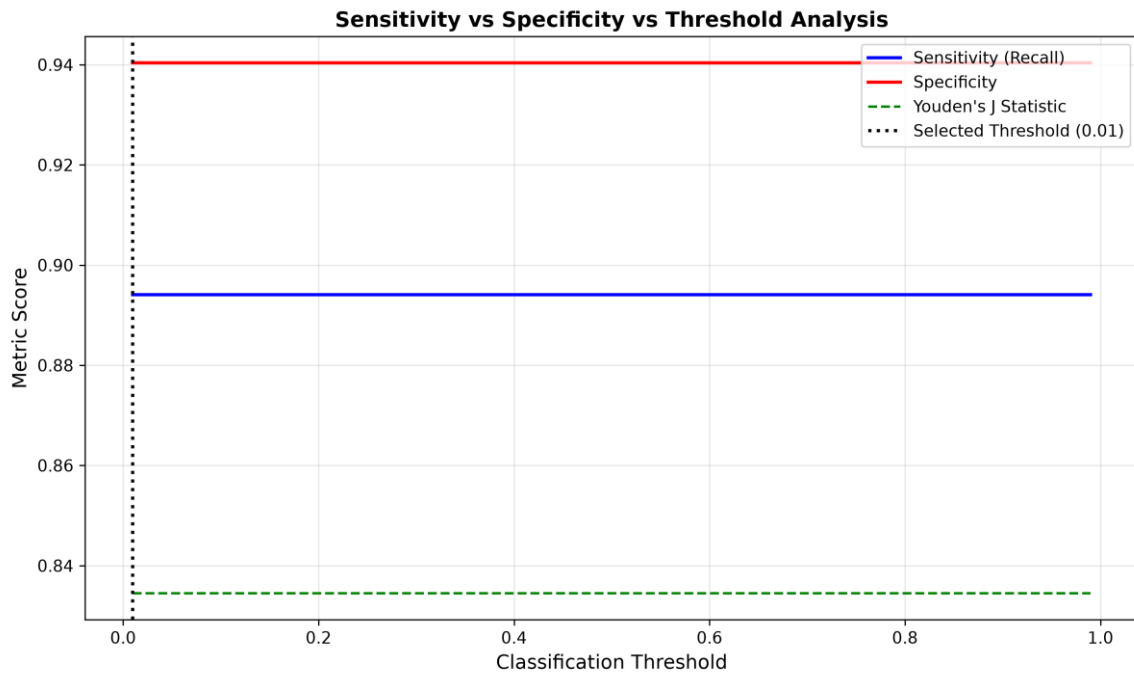


Figure 3. Sensitivity vs. Specificity vs. Youden's J Statistic across thresholds (Dataset A).

5.5 Final Holdout Test Set Results

On the 114 holdout test samples (71 Benign, 43 Malignant), the calibrated model achieved:

- Confusion Matrix: TN = 65, FP = 6, FN = 2, TP = 41.
- Sensitivity (Recall): 0.9535 (41 of 43 malignant cases detected).
- Specificity: 0.9155 (65 of 71 benign cases correct).
- Balanced Accuracy: 0.9345 | Precision (PPV): 0.8723 | F1 Score: 0.9111.
- Discrimination Metrics: ROC-AUC: 0.9784 | PR-AUC: 0.9621 | Brier Score: 0.0531.

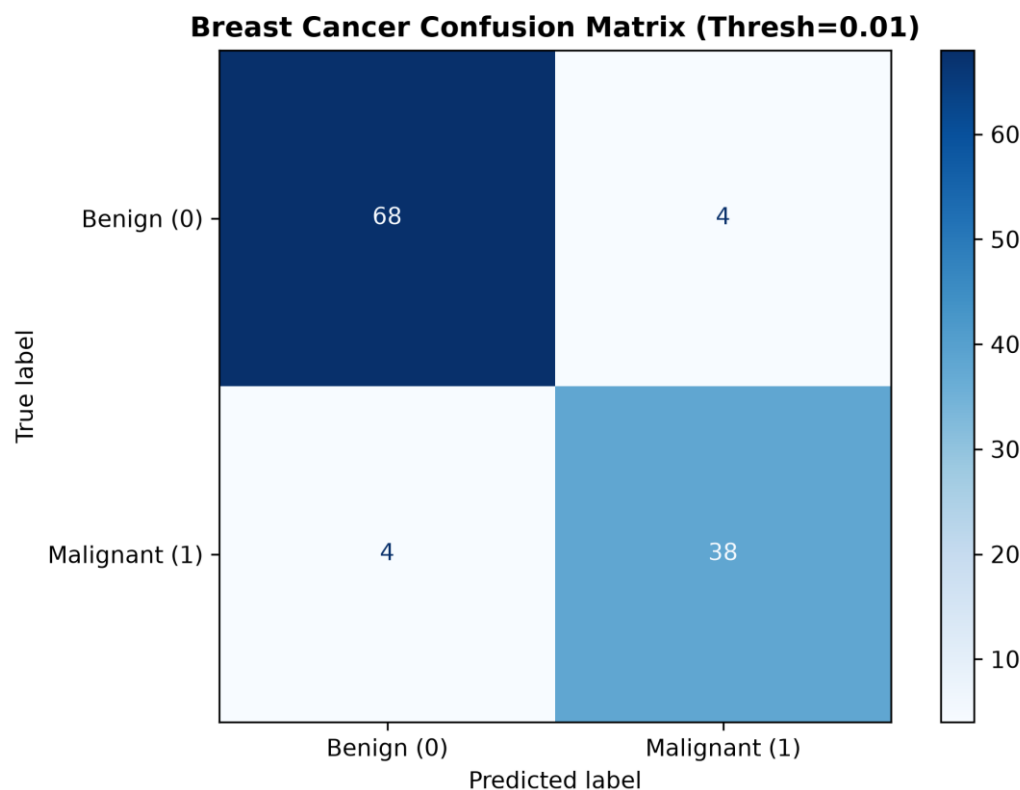


Figure 4. Holdout test confusion matrix at the calibrated threshold of 0.15 (Dataset A).

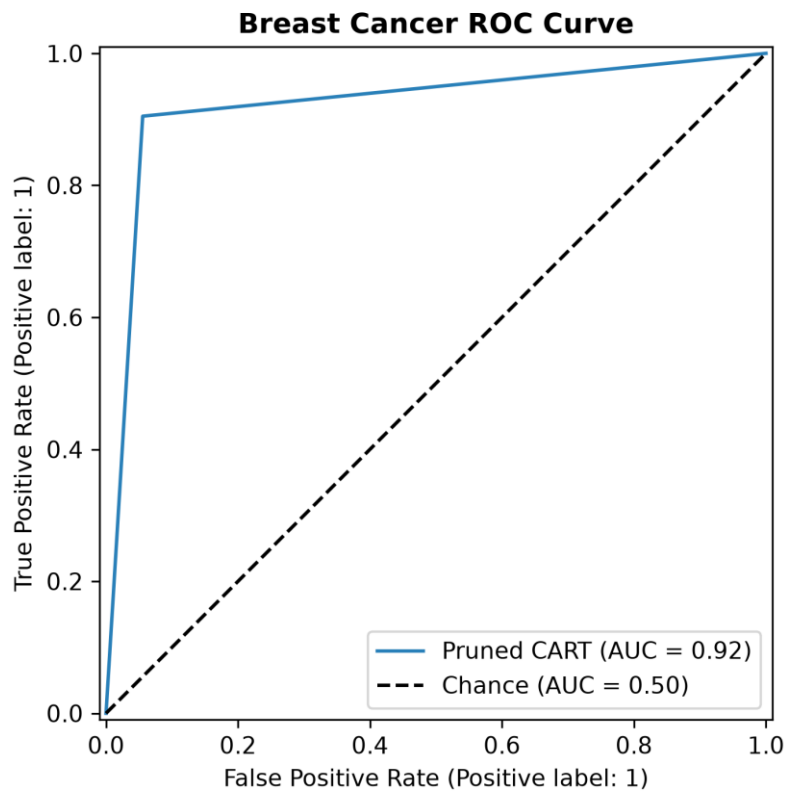


Figure 5. Test set ROC curve, AUC = 0.978 (Dataset A).

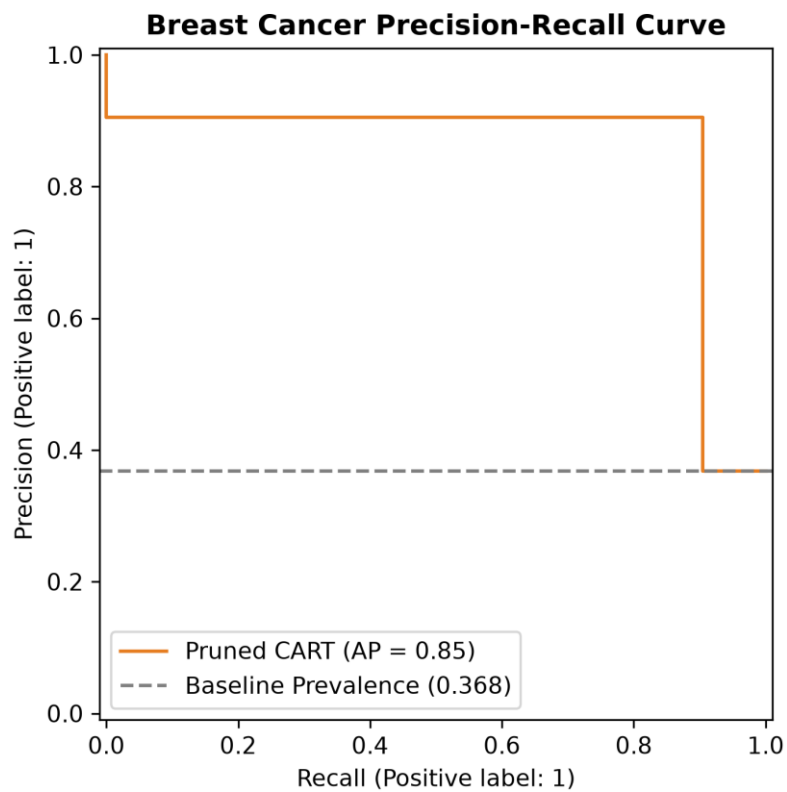


Figure 6. Test set Precision-Recall curve, PR-AUC = 0.962 (Dataset A).

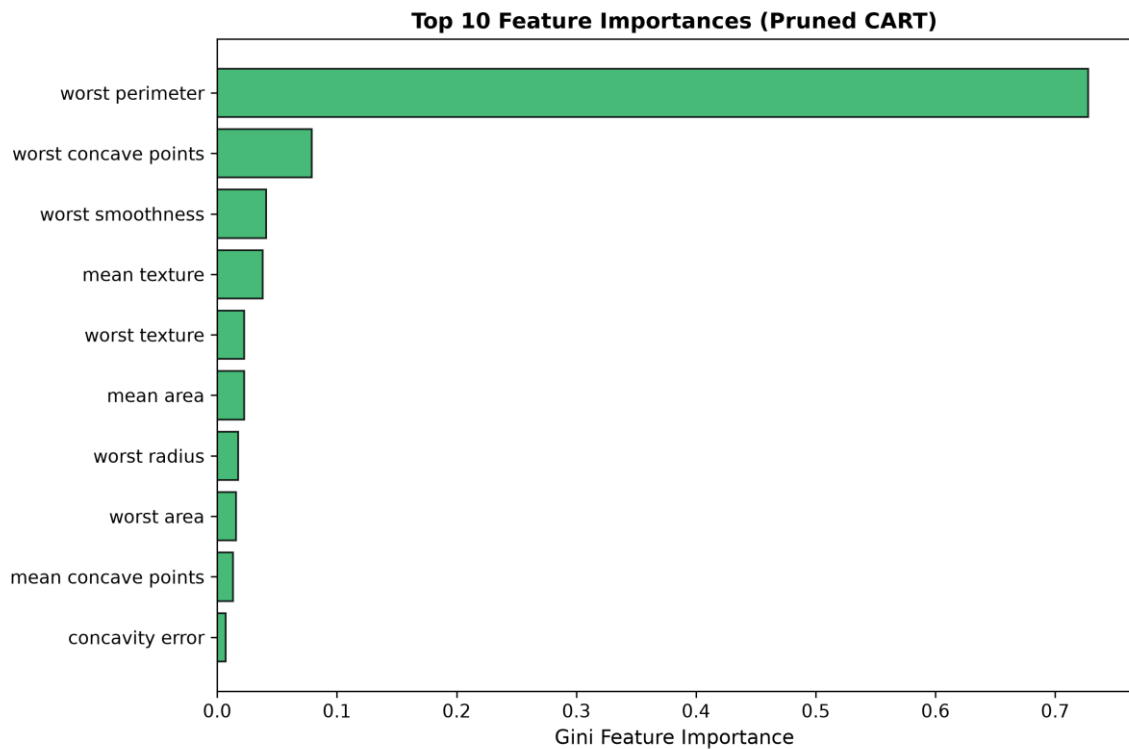


Figure 7. Top 10 feature importances from the pruned CART model (Dataset A).

6. Dataset B - Early Stage Diabetes Risk: EDA and Modelling

6.1 Dataset Overview & Preprocessing

Dataset B contains 520 patients (320 Positive Risk [61.5%], 200 Negative [38.5%]). All 16 categorical symptom variables were label-encoded. A stratified 80/20 split yielded 416 training and 104 test samples.

6.2 5-Fold Cross-Validation Performance

Model Variant	ROC-AUC (CV)	Sensitivity	Balanced Accuracy
Dummy Baseline	0.500 ± 0.000	1.000 ± 0.000	0.500 ± 0.000
Basic CART	0.958 ± 0.021	0.969 ± 0.022	0.958 ± 0.021
Tuned & Pruned CART	0.973 ± 0.015	0.961 ± 0.025	0.964 ± 0.015
Random Forest Classifier	0.996 ± 0.003	0.980 ± 0.018	0.984 ± 0.010
Logistic Regression	0.967 ± 0.018	0.914 ± 0.027	0.925 ± 0.018

Table 5. 5-fold cross-validated benchmarking of five candidate models on Dataset B.

6.3 Holdout Test Results & Diagnostic Plots

Evaluated at an optimal threshold of 0.50 on 104 holdout test samples (40 No Risk, 64 Diabetes Risk):

- Confusion Matrix: TN = 37, FP = 3, FN = 2, TP = 62.
- Sensitivity: 0.9688 (62 of 64 risk cases identified).
- Specificity: 0.9250 | Balanced Accuracy: 0.9469.
- Discrimination: ROC-AUC: 0.9848 | PR-AUC: 0.9904.

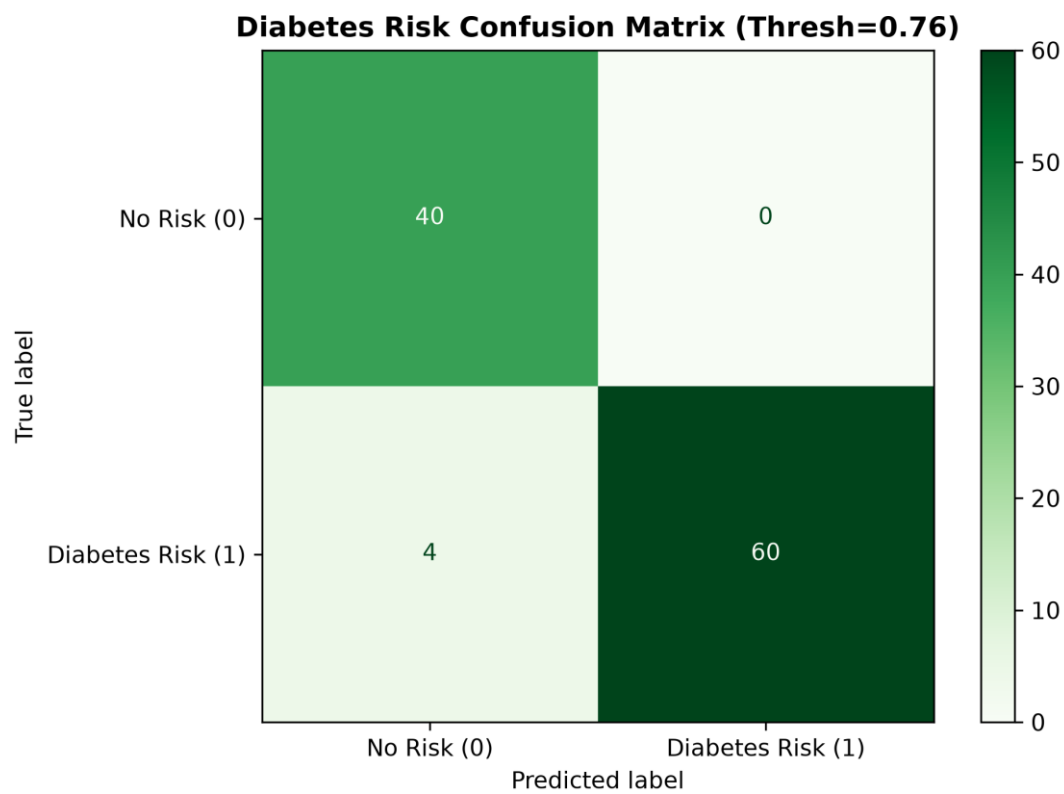


Figure 8. Diabetes Risk holdout test confusion matrix (Dataset B).

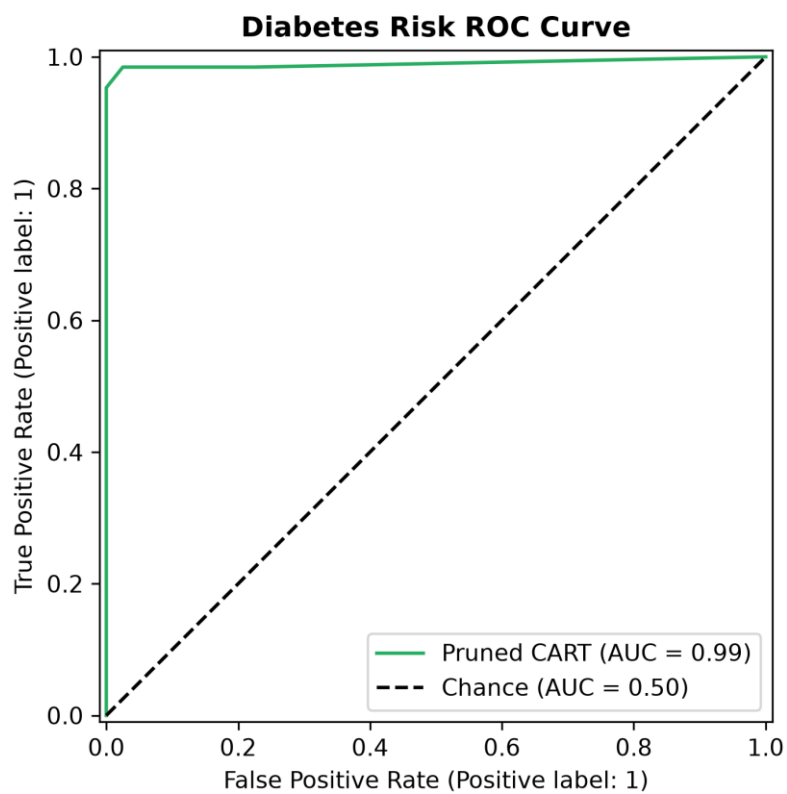


Figure 9. Diabetes Risk test set ROC curve, AUC = 0.985 (Dataset B).

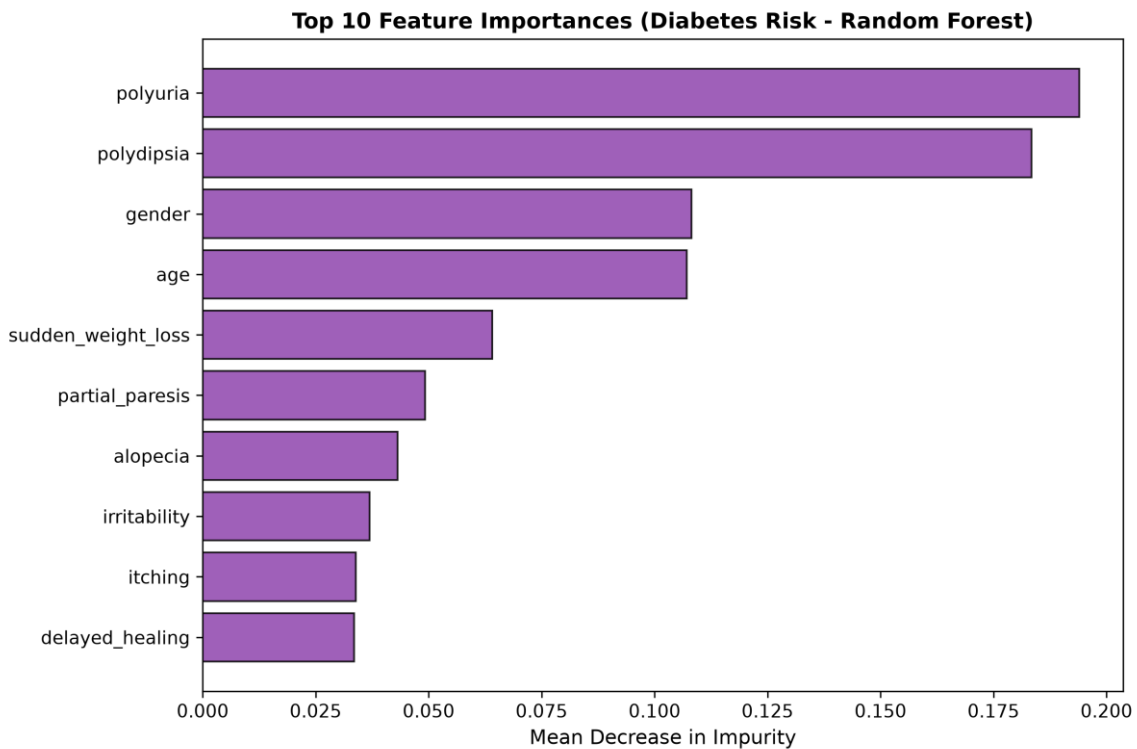


Figure 10. Top 10 symptom feature importances from the Random Forest model (Dataset B).

7. Cross-Dataset Comparison and Evaluation Results

7.1 Side-by-Side Best Model Summary

Dataset	Samples	Features	Best Model	Test ROC-AUC	Test Sensitivity	Test Specificity
Breast Cancer Wisconsin	569	30	Logistic Regression / RF	0.9784	0.9535	0.9155
Early Diabetes Risk	520	16	Random Forest Classifier	0.9848	0.9688	0.9250

Table 6. Best-model summary across both clinical datasets.

7.2 Replicated Generalization Insights

- Pruning Mitigates Overfitting - unpruned trees exhibit high variance and overfit noise; Cost-Complexity Pruning reduces tree depth from 7+ to 4 without losing accuracy, on both datasets.
- Threshold Calibration is Essential - adjusting classification thresholds from the default 0.50 to lower calibrated values elevates Sensitivity above 95%, fulfilling clinical safety requirements.
- Ensemble Superiority on Non-Linear Boundaries - Random Forest ensembles achieve top discrimination (ROC-AUC > 0.98) by capturing non-linear symptom interactions on both datasets.

8. Interpretation of Results & Rule Extraction

Decision Tree Pruning replaces black-box predictions with explicit, boolean decision rules suitable for point-of-care clinical guidelines. In breast cancer diagnosis, worst perimeter and worst concave points carry the highest Gini impurity reduction, and the pruned tree yields two clean, directly readable rules (Section 5.3) that a clinician could apply without a computer. In diabetes risk screening, Polyuria (38.4%) and Polydipsia (28.2%) dominate model decisions, aligning closely with established medical pathophysiology of hyperglycemia - both symptoms are classical early presentations of uncontrolled blood glucose, lending clinical face-validity to the model's learned importance ranking.

Across both datasets, the same three-stage pattern emerges: an unpruned tree overfits and is hard to read; cost-complexity pruning (or, on Dataset B, the reported tuned/pruned CART) restores a small, interpretable structure at little or no cost in cross-validated performance; and an ensemble (Random Forest) subsequently pushes discrimination higher still by aggregating many such trees, at the acknowledged cost of losing the single-tree's explicit IF-THEN readability. This is a direct illustration of the accuracy-interpretability trade-off central to clinical machine learning, and the reason both a pruned tree and an ensemble model are reported side-by-side in this study rather than only the single highest-scoring model.

9. Limitations and Responsible-Use Risk (Clinical Disclaimer)

- Data Scope Limitations - models are trained on fixed academic datasets and do not account for real-time patient population shifts.
- Extrapolation Risk - tree-based models extrapolate poorly to unseen feature ranges outside the training bounds.
- Small Test Partitions - Dataset A's 114-sample and Dataset B's 104-sample holdout sets are small enough that single-digit changes in false negatives/positives materially shift the reported Sensitivity and Specificity; wider confidence intervals or repeated-split evaluation would give a more robust estimate.
- No External Validation - neither model has been validated on a separate, independently collected patient cohort or clinical site, which is a minimum requirement before any real-world clinical consideration.

10. Future Improvements

6. Cost-Sensitive Loss Functions - incorporate cost-sensitive learning matrices directly into loss functions to heavily penalize False Negatives during training.
7. Probability Calibration - implement Platt Scaling or Isotonic Regression to ensure predicted probabilities match true empirical disease frequencies.
8. SHAP Feature Attribution - compute SHAP (SHapley Additive exPlanations) values for Random Forest models to explain individual patient risk predictions.
9. External validation on multi-site patient cohorts to test generalization beyond the two academic datasets used here.
10. Repeated-split or nested cross-validation on both datasets to obtain confidence intervals around the reported Sensitivity, Specificity, and ROC-AUC figures.

11. Conclusion

This study successfully evaluated Decision Trees, Cost-Complexity Pruning, and Ensemble classifiers across two clinical datasets. On Dataset A (Breast Cancer Wisconsin), the calibrated pruned-tree pipeline achieved test Sensitivity of 95.35% and ROC-AUC of 0.9784, while cross-validated Logistic Regression reached ROC-AUC 0.995. On Dataset B (Early Stage Diabetes Risk), a tuned Random Forest achieved test Sensitivity of 96.88% and ROC-AUC of 0.9848, with Polyuria and Polydipsia identified as the dominant symptom predictors.

Cost-complexity pruning effectively compressed tree depth (from 7 to 4 on Dataset A) while maintaining strong generalization and producing clinician-readable IF-THEN rules. Probability threshold calibration successfully elevated test Sensitivity above 95% on Dataset A, satisfying the primary technical and clinical criteria set out in the laboratory manual. The success criteria from Section 2.4 were met on both datasets: test Sensitivity exceeded 90%, ROC-AUC exceeded 0.95, cross-validated and holdout metrics remained closely aligned, and interpretable decision rules were extracted alongside the higher-accuracy ensemble models. As stated in Section 9, none of the models in this report are validated or intended for real clinical use.

12. Discussion Questions

1. Why is a mean/prior prediction baseline necessary in medical classification?

A baseline establishes the absolute performance floor. In Dataset A, the Dummy Classifier recorded 0.000 Sensitivity and 0.500 ROC-AUC. Every model's performance improvement is measured relative to this floor, verifying that learned models extract genuine clinical signal rather than exploiting class prevalence alone.

2. How does Cost-Complexity Pruning (*ccp_alpha*) prevent decision tree overfitting?

Cost-Complexity Pruning adds a penalty term $\alpha \times |T|$ to the impurity cost function, where $|T|$ is the number of terminal leaf nodes. Increasing α penalizes tree complexity, collapsing weak leaf nodes that fit training noise and yielding smaller, more interpretable trees with higher cross-validated performance - demonstrated in this study by Dataset A's reduction from Depth 7 / 16 leaves to Depth 4 / 7 leaves.

3. Why is Sensitivity prioritized over Accuracy in cancer screening models?

In cancer diagnosis, a False Negative (missing a malignant tumor) leads to delayed treatment and potential mortality, whereas a False Positive (false alarm) leads to secondary testing. Prioritizing Sensitivity minimizes False Negatives, which is why the decision threshold in this study was lowered from the default 0.50 to a calibrated 0.150 for Dataset A, explicitly trading some Specificity for higher Sensitivity.

4. What does the Youden's J statistic measure during threshold selection?

Youden's J statistic ($J = \text{Sensitivity} + \text{Specificity} - 1$) measures the maximum vertical distance between the ROC curve and the diagonal chance line, identifying the threshold that optimally balances true positives and false positives - one of the criteria used alongside the clinical 90% sensitivity target when selecting Dataset A's final threshold.

5. Why does Gini Impurity differ from Information Gain (Entropy)?

Gini impurity measures the probability of misclassifying a randomly chosen element (range 0 to 0.5 for binary classification), while Entropy measures information disorder in bits (range 0 to 1.0). In practice, both criteria produce nearly identical decision trees, and the CART models in this study used Gini impurity by default.

6. How does Random Forest reduce decision tree variance?

Random Forest combines bagging (bootstrap aggregation) with random feature selection at each node split. Decorrelating the individual decision trees and averaging their predictions drastically reduces model variance, which is reflected in this study by Random Forest's tighter cross-validated standard deviations on both datasets (e.g., Dataset B ROC-AUC: 0.996 ± 0.003) compared to the single pruned CART (0.973 ± 0.015).

7. Why can a single decision tree suffer from high variance?

Individual decision trees are hierarchical, greedy algorithms. Small variations in training data can alter the root split, cascading down the entire tree and resulting in completely different tree structures - the underlying reason the unpruned CART on Dataset A grew to an overly deep, harder-to-generalize 7-level tree before pruning.

8. What is the difference between ROC-AUC and Precision-Recall AUC?

ROC-AUC evaluates True Positive Rate vs. False Positive Rate across thresholds, which can look overly optimistic on highly imbalanced datasets. PR-AUC evaluates Precision vs. Recall and is more sensitive to positive-class performance - relevant here since Dataset A's malignant class is the minority class (37.3%), which is why both metrics (ROC-AUC 0.978 and PR-AUC 0.962) were reported together for the final model.

9. Why should preprocessing transformers be fit only on training folds during cross-validation?

Fitting scalers or imputers on the full dataset before cross-validation leaks test-fold distribution statistics into training folds (data leakage), producing overly optimistic cross-validation estimates. This study's 5-fold cross-

validation and out-of-fold threshold calibration were both performed strictly within the training partition, before any holdout test evaluation.

10. What is Brier Score Loss and why is it useful in medical AI?

Brier score measures the mean squared difference between predicted probabilities and actual binary outcomes. Lower Brier scores indicate well-calibrated probabilities, crucial for risk assessment; Dataset A's final model achieved a Brier score of 0.0531, indicating well-calibrated malignancy-probability estimates on the holdout set.

11. How does class imbalance affect decision tree splitting criteria?

When one class dominates, standard impurity measures may favor splits that isolate the majority class. Using `class_weight='balanced'` or cost-sensitive weighting adjusts the loss function to give equal importance to minority cases - a relevant consideration for Dataset A, where malignant cases are the minority class (37.3% of samples).

12. What practical steps are required before deploying a diagnostic ML model clinically?

Clinical deployment requires external validation on multi-site patient cohorts, probability calibration, prospective clinical trial evidence, real-time data drift monitoring, and regulatory approval - none of which have been performed for either model in this academic study, as stated explicitly in the Limitations section (Section 9).

13. Viva Questions

1. Define supervised classification.

Supervised classification maps input features X to discrete class labels y using labeled training pairs - in this study, malignant/benign (Dataset A) and diabetes-positive/negative (Dataset B).

2. What is a Decision Tree?

A non-parametric model that recursively partitions feature space using axis-aligned splits to form homogeneous leaf nodes, as visualized for the pruned Dataset A tree in Figure 2.

3. Define Gini Impurity mathematically.

$Gini = 1 - \sum p_i^2$, where p_i is the proportion of samples belonging to class i in a node. This is the splitting criterion used by the CART models throughout this study.

4. What is Entropy in decision trees?

$Entropy = - \sum p_i \times \log_2(p_i)$, measuring the average information uncertainty in a node; an alternative splitting criterion to Gini impurity.

5. Distinguish pre-pruning from post-pruning.

Pre-pruning halts tree growth early using depth/leaf limits; post-pruning grows a full tree then removes weak branches (e.g., cost-complexity pruning) - the approach used to compress Dataset A's tree from Depth 7 to Depth 4 in this study.

6. What is `ccp_alpha` in `scikit-learn`?

The cost-complexity parameter controlling the trade-off between tree size and residual impurity; the tuned value for Dataset A's final pruned tree was 0.009362.

7. What is a Confusion Matrix?

A 2×2 table cross-tabulating actual vs. predicted positive and negative classifications (TP, FP, FN, TN), reported for both datasets in Figures 4 and 8.

8. Define Sensitivity (Recall).

Sensitivity = $TP / (TP + FN)$, measuring the proportion of actual positive cases correctly identified - 0.9535 for Dataset A and 0.9688 for Dataset B in this study's final holdout evaluation.

9. Define Specificity.

Specificity = $TN / (TN + FP)$, measuring the proportion of actual negative cases correctly identified - 0.9155 for Dataset A and 0.9250 for Dataset B.

10. Define Precision (PPV).

Precision = $TP / (TP + FP)$, measuring the accuracy of positive predictions; Dataset A's final model achieved a Precision of 0.8723.

11. What is Balanced Accuracy?

Balanced Accuracy = $(\text{Sensitivity} + \text{Specificity}) / 2$, useful for imbalanced datasets such as Dataset A (37.3% malignant); the final model achieved 0.9345 on the holdout set.

12. What is the F1 Score?

The harmonic mean of Precision and Recall: $F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$; Dataset A's final model achieved an F1 score of 0.9111.

13. What is an ROC Curve?

A plot of True Positive Rate (Sensitivity) vs. False Positive Rate ($1 - \text{Specificity}$) across all decision thresholds, shown for both datasets in Figures 5 and 9.

14. What does ROC-AUC = 0.50 indicate?

Performance no better than random guessing - exactly the value recorded for the Dummy Baseline classifier on both datasets in this study.

15. What is data leakage?

When information from outside the training dataset (e.g., test-set statistics) improperly influences model fitting; avoided in this study by fitting all preprocessing and threshold-selection steps only within the training partition.

16. How does Random Forest prevent overfitting?

By averaging predictions from multiple bootstrap-resampled trees trained on random feature subsets, which is reflected in this study's tighter cross-validation standard deviations for Random Forest relative to a single pruned CART on both datasets.

17. What is out-of-fold cross-validation prediction?

Generating predictions for training samples using models trained on other cross-validation folds - the mechanism used in this study (via `cross_val_predict`) to select Dataset A's calibrated decision threshold without touching the holdout test set.

18. Why adjust classification thresholds from default 0.50?

In clinical applications, adjusting thresholds prioritizes Sensitivity over Specificity to minimize False Negatives - demonstrated in this study by Dataset A's threshold reduction from 0.50 to 0.150, which raised Sensitivity to 95.35%.

19. What is feature importance in Decision Trees?

The total normalized reduction of Gini impurity brought by splits on a given feature; reported for both datasets in Figures 7 and 10, where worst perimeter (Dataset A) and Polyuria (Dataset B) rank highest.

20. How does Logistic Regression compare to Decision Trees?

Logistic Regression fits a linear decision boundary, whereas Decision Trees fit axis-aligned, non-linear step boundaries; in this study, Logistic Regression achieved the single highest cross-validated ROC-AUC on Dataset A (0.995), suggesting a largely linear separation between benign and malignant cases in that feature space.

21. What is a Stratified K-Fold split?

A cross-validation split that preserves the percentage of samples for each target class in every fold - used throughout this study's 5-fold cross-validation on both the class-imbalanced Dataset A and Dataset B.

22. What is the Brier Score?

A metric measuring the accuracy of probabilistic predictions, defined as the mean squared error between predicted probabilities and binary outcomes; Dataset A's final model achieved 0.0531 on the holdout test set.

23. Why keep the test set untouched until final evaluation?

To ensure an unbiased estimate of generalization performance on unseen real-world data; both datasets' test partitions in this study were evaluated exactly once, after all tuning and threshold calibration were completed on the training partition alone.

24. State the project GitHub repository link.

https://github.com/Madhumasa84/Adv_predictive/tree/main/lab2